

5.1 Linear independence of stacked vectors. Consider the stacked vectors

$$c_1 = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}, \dots, c_k = \begin{bmatrix} a_k \\ b_k \end{bmatrix},$$

where $a_1, \dots, a_k$ are n -vectors and $b_1, \dots, b_k$ are m -vectors.

- (a) Suppose $a_1, \dots, a_k$ are linearly independent. (We make no assumptions about the vectors $b_1, \dots, b_k$.) Can we conclude that the stacked vectors $c_1, \dots, c_k$ are linearly independent?
- (b) Now suppose that $a_1, \dots, a_k$ are linearly dependent. (Again, with no assumptions about $b_1, \dots, b_k$.) Can we conclude that the stacked vectors $c_1, \dots, c_k$ are linearly dependent?

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

Block matrix.

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$\nearrow$
 $\underline{x}$

$$= \begin{pmatrix} A x_1 + B x_2 \\ C x_1 + D x_2 \end{pmatrix}$$

$$\begin{array}{c|c} A & B \\ \hline C & D \end{array}$$

$$(a) \quad C = \begin{bmatrix} A \\ B \end{bmatrix}$$

$$C \underline{x} = \begin{bmatrix} A \\ B \end{bmatrix} \underline{x} = \begin{bmatrix} A \underline{x} \\ B \underline{x} \end{bmatrix}$$

$$\underline{C \underline{x}} = \underline{0} \Rightarrow \underbrace{A \underline{x} = 0}_{\downarrow} \text{ and } \underbrace{B \underline{x} = 0}$$

$$\Rightarrow \underline{x} = 0$$

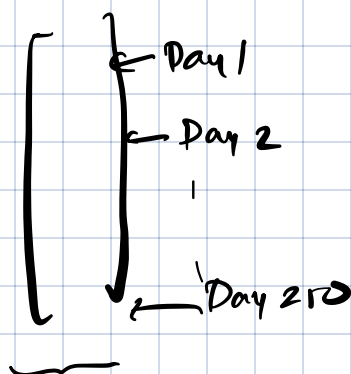
$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \sqrt{2} & -\pi \\ \sqrt{3} & -e \\ \sqrt{15} & -2i \end{pmatrix}$$

$$C = \begin{bmatrix} A & B \\ \underline{I} & \underline{I} \end{bmatrix}$$

5.2 *A surprising discovery.* An intern at a quantitative hedge fund examines the daily returns of around 400 stocks over one year (which has 250 trading days). She tells her supervisor that she has discovered that the returns of one of the stocks, Google (GOOG), can be expressed as a linear combination of the others, which include many stocks that are unrelated to Google (say, in a different type of business or sector).

Her supervisor then says: "It is overwhelmingly unlikely that a linear combination of the returns of unrelated companies can reproduce the daily return of GOOG. So you've made a mistake in your calculations."

Is the supervisor right? Did the intern make a mistake? Give a very brief explanation.



M.4 Recall the definition and first two derivatives of a third-order polynomial:

$$f(x) = a + bx + cx^2 + dx^3$$

$$f'(x) = b + 2cx + 3dx^2$$

$$f''(x) = 2c + 6dx$$

- (a) Write the matrix that computes the first three derivatives of the above polynomial for a given x , i.e. give A such that

$$\begin{bmatrix} f(x) \\ f'(x) \\ f''(x) \end{bmatrix} = A \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \end{bmatrix}$$

$$A = \begin{bmatrix} a & b & c & d \\ b & 2c & 3d & 0 \\ 2c & 6d & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a & b & 2c \\ b & 2c & 3d \\ c & 3d & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

- (b) Is it possible for the *columns* of A to be linearly independent? If yes, give an example of conditions on a, b, c, d that make the columns independent. If no, provide a brief justification.

No
 $d \neq 0$ or $c \neq 0$

- (c) Is it possible for the *rows* of A to be linearly independent? If yes, give an example of conditions on a, b, c, d that make the rows independent. If no, provide a brief justification.

$$\begin{bmatrix} a & b & c & d \\ b & 2c & 3d & 0 \\ 2c & 6d & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

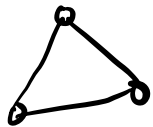
$$\underbrace{bx_1 + 2cx_2 + 6dx_3 = 0}$$

$$\underbrace{cx_1 + 3x_2 = 0}$$

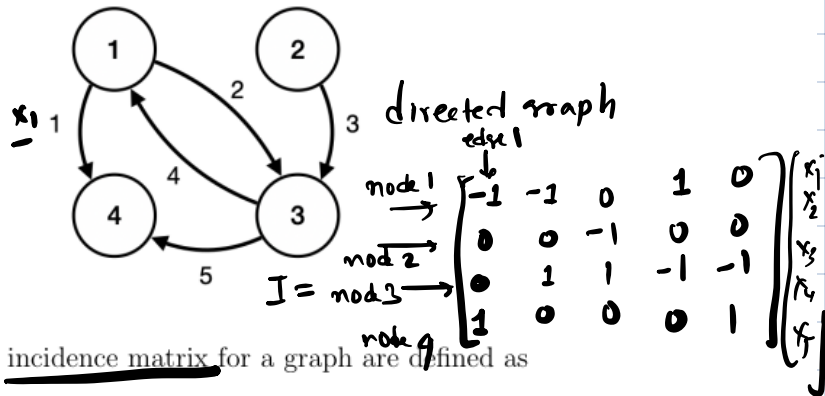
$$\underbrace{dx_1 = 0}$$

M.4 Consider the graph below, with labeled nodes and edges.

Graph matrices



$I_x =$



Recall that the entries of the incidence matrix for a graph are defined as

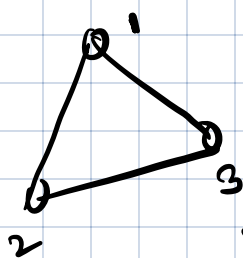
$$A_{ij} = \begin{cases} 1 & \text{edge } j \text{ points to node } i \\ -1 & \text{edge } j \text{ points from node } i \\ 0 & \text{otherwise} \end{cases} = \begin{bmatrix} -x_1 - x_2 + x_3 \\ -x_3 \\ x_2 + x_3 - x_4 - x_5 \end{bmatrix}$$

- Write the incidence matrix for this graph.
- Recall the vector of net node flows $y = Ax$, where A is the incidence matrix and x are the flows along each edge. Suppose we have a pre-specified y and would like to find edge flows x that produce it. Can we do so using a left inverse of A ? Why or why not?
- Now suppose we have a required value for the first flow, $x_1 = 1$. Using this condition, rewrite the system from (b) as $\tilde{y} = \tilde{A}\tilde{x}$, where $\tilde{x} = (x_2, \dots, x_k)$, where k is the number of edges. Does $\tilde{A}$ have a left inverse? Why or why not?

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & -1 & -1 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Adjacency matrix of a graph

square matrix



node 1 node 2 node 3

node 1 →

node 2 →

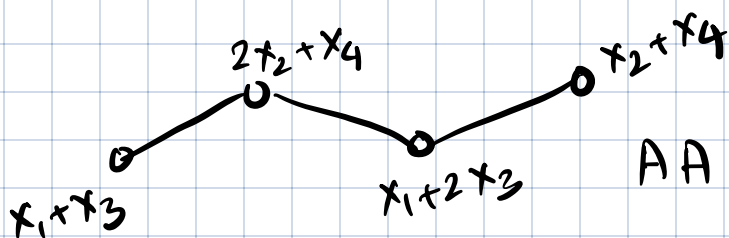
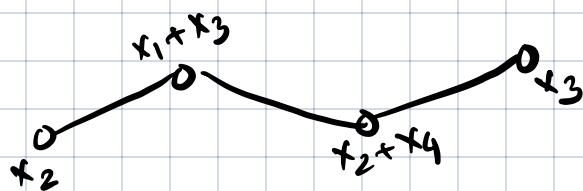
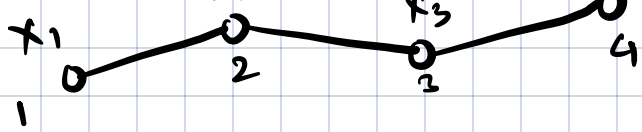
node 3 →

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 + x_3 \\ x_1 + x_3 \\ x_2 + x_1 \end{bmatrix}$$

x_2

x_4



$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$AA^T = ?$$

Exercise: Let I be an incidence matrix of a graph.
What is $\underbrace{II^T}$? Can we interpret it operationally?

Gram matrix: the gram matrix of A Sometimes.
is written as $\underbrace{A^T A}_{n \times n}$ ($\underbrace{AA^T}$)

$$\underbrace{\begin{bmatrix} \text{row 1} \\ \text{row 2} \\ \text{row 3} \end{bmatrix}}_{A^T} \underbrace{\begin{bmatrix} \text{col 1} & \text{col 2} & \text{col 3} & 0 \end{bmatrix}}_A$$

$(i,j)^{\text{th}}$ entry of $A^T A$ is the inner product of i^{th} column of A with j^{th} column of A .

$$BA = I$$

$$AB = I$$

$$\underbrace{BA} = I$$

$$\begin{bmatrix} \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} \vdots & \vdots & \vdots & \vdots \end{bmatrix} = I$$

$$\begin{bmatrix} \begin{bmatrix} \uparrow \\ \underbrace{\quad}_A \end{bmatrix} \\ \left[\begin{array}{c} \uparrow \\ \underbrace{\quad} \end{array} \right] \end{bmatrix} \neq I$$

, 2